ALGEBRA 1 · UNIT 11: QUADRATIC FUNCTIONS AND EQUATIONS

Launch Lab

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run a launch lab for the science fair: they graph and analyze projectile height functions, find maximum heights and landing times by factoring, the quadratic formula, and completing the square, use the discriminant to test claims, and design a launch that follows the park's height rules.

Projectiles are the classic quadratic context because every key feature has meaning: the vertex is the highest point, the positive zero is the landing time, and the domain stops at the ground. Each level uses the solving methods appropriate to it, and Honors students fit a model from data.

Standards Addressed

- F-IF.4 Interpret key features of quadratic graphs
- A-REI.4 Solve quadratic equations (factoring, square roots, completing the square, formula)
- F-IF.8a Use factored and vertex forms to reveal zeros and extreme values
- F-LE.3 Compare linear, exponential, and quadratic models

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: if possible, launch a stomp rocket and time it. Part 1: tables, graphs, key features.
Day 2	Part 2: solve for heights and landing times.
Day 3	Part 3: design a launch and write the report.

Materials

- Student handout
- Graphing calculators or Desmos
- Optional: stomp rockets or water balloons and a stopwatch for a real launch

How the Three Versions Differ

Level	What changes
CP2	Rockets launched from the ground, factorable equations, vertex from the axis of symmetry, and a design with v as a multiple of 16.
CP1	Launch from a platform, quadratic formula, vertex form, discriminant claims, and comparing linear, exponential, and quadratic tables.
Honors	Fits a quadratic from data, completes the square, exact radical answers, Mars gravity, and an inequality on launch speed for a height rule.

Watch For (Common Misconceptions)

- Thinking the graph shows the path of the rocket (it shows height vs. time).
- Keeping the negative root as a landing time.
- Using t = −b ÷ 2a but forgetting to substitute to get the height.

Extension Ideas

- Run real launches and fit a model with Desmos regression.
- Add horizontal distance for a full trajectory.
- Compare Earth, Moon, and Mars gravity.

Answer Key – CP2

Part 1: Rocket on the Ground

#	Task	Answer
1	Complete the table.	0, 48, 64, 48, 0
2	Graph the function on the grid.	Parabola opening down through (0, 0), (1, 48), (2, 64), (3, 48), (4, 0).
3	Find the vertex, the axis of symmetry, and the zeros. What does each mean for the rocket?	Vertex (2, 64): highest point, 64 ft at 2 s. Axis $t = 2$. Zeros $t = 0$ (launch) and $t = 4$ (landing).
4	What are a reasonable domain and range for this launch?	Domain $0 \le t \le 4$; range $0 \le h \le 64$

Part 2: Heights & Landings

#	Task	Answer
5	Factor $h(t) = -16t^2 + 64t$. Use the factored form to explain when the rocket is on the ground.	$h(t) = -16t(t - 4)$. It is 0 when $t = 0$ or $t = 4$: at launch and when it lands.
6	When is the rocket 48 feet high? Solve $-16t^2 + 64t = 48$.	$-16t^2 + 64t - 48 = 0 \rightarrow t^2 - 4t + 3 = 0 \rightarrow (t - 1)(t - 3) = 0 \rightarrow t = 1$ s and $t = 3$ s (going up and coming down)
7	A water balloon is tossed up from a window: $h(t) = -16t^2 + 16t + 32$. Factor and find when it hits the ground.	$-16(t^2 - t - 2) = -16(t - 2)(t + 1) = 0 \rightarrow t = 2$ s (reject -1)
8	Find the water balloon's maximum height. Use the axis of symmetry $t = -b \div (2a)$.	$t = -16 \div (2 \cdot -16) = ½$; $h(½) = -4 + 8 + 32 = 36$ ft

Part 3: Design a Launch

#	Task	Answer
9	Choose a launch speed v that is a multiple of 16. Write $h(t) = -16t^2 + vt$.	Answers vary. Example: $v = 80$.
10	Find your rocket's maximum height and landing time. Does it follow the park rule?	Example $v = 80$: vertex at $t = 80 \div 32 = 2.5$ s, $h = -100 + 200 = 100$ ft (allowed, exactly 100). Lands at $t = 5$ s. $v = 96$ gives 144 ft (too high).
11	Graph your launch and the original rocket on one grid. Compare the two.	Answers vary.

Answer Key – CP1

Part 1: Rocket from a Platform

#	Task	Answer
1	Find the vertex and explain its meaning.	$t = -64 \div (-32) = 2$; $h(2) = -64 + 128 + 4 = 68$. Vertex (2, 68): max height 68 ft at 2 s.
2	Write h(t) in vertex form and check that it matches your vertex.	$h(t) = -16(t - 2)^2 + 68$. Expanding: $-16t^2 + 64t - 64 + 68 = -16t^2 + 64t + 4$ ✓
3	Use the quadratic formula to find when the rocket lands. Round to the nearest hundredth.	$t = (-64 \pm \sqrt{(4096 + 256)}) \div (-32) = (-64 \pm \sqrt{4352}) \div (-32) \rightarrow t \approx 4.06$ s (reject ≈ −0.06)
4	Rocket B follows $h(t) = -16(t - 1.5)^2 + 40$. Which rocket goes higher? Which is in the air longer? Justify.	Platform rocket: 68 ft > 40 ft, so it goes higher. Rocket B lands when $(t - 1.5)^2 = 2.5 \rightarrow t \approx 3.08$ s, so the platform rocket (4.06 s) is in the air longer.

Part 2: Claims & Comparisons

#	Task	Answer
5	A visitor claims the platform rocket will reach 80 feet. Use the discriminant to decide.	$-16t^2 + 64t - 76 = 0$; $b^2 - 4ac = 4096 - 4(-16)(-76) = 4096 - 4864 = -768 < 0$. No: it never reaches 80 ft.
6	When is the platform rocket exactly 60 feet high? Give answers to the nearest hundredth.	$-16t^2 + 64t - 56 = 0 \rightarrow 2t^2 - 8t + 7 = 0 \rightarrow t = (8 \pm \sqrt{8}) \div 4 \approx$ 1.29 s and 2.71 s
7	A water balloon follows $h(t) = -16t^2 + 16t + 32$. Solve by factoring to find when it lands.	$-16(t - 2)(t + 1) = 0 \rightarrow t = 2$ s
8	Classify each model in the table as linear, exponential, or quadratic. Explain using differences or ratios.	A: linear (adds 4). B: exponential (multiplies by 2). C: quadratic (first differences 3, 5, 7; second differences 2).

Part 3: Design a Launch

#	Task	Answer
9	Choose a launch speed v and write $h(t) = -16t^2 + vt + 4$.	Answers vary.
10	Find the vertex. Show that it follows the rule.	Max $= 4 + v^2 \div 64$. Example v = 72: t = 2.25, h = 85 ft ✓
11	Find the landing time with the quadratic formula and present your launch with a graph.	Example v = 72: $-16t^2 + 72t + 4 = 0 \rightarrow t \approx 4.55$ s

Answer Key – Honors

Part 1: Model from Data

#	Task	Answer
1	Use $h(t) = at^2 + bt + c$ and the three data points to write and solve a system for a, b, and c.	$c = 5$; $a + b = 64$; $4a + 2b = 96 \rightarrow 2a = -32 \rightarrow a = -16$, $b = 80$. $h(t) = -16t^2 + 80t + 5$
2	Complete the square to write the model in vertex form. State the maximum height and when it occurs.	$-16(t^2 - 5t) + 5 = -16(t - 2.5)^2 + 100 + 5 = -16(t - 2.5)^2 + 105$. Max 105 ft at 2.5 s.
3	Find the exact landing time in simplest radical form, then approximate it.	$-16(t - 2.5)^2 + 105 = 0 \rightarrow (t - 2.5)^2 = 105 \div 16 \rightarrow t = 2.5 + \frac{1}{4}\sqrt{105} \approx 5.06$ s
4	Show that the data have constant second differences. Why does that confirm a quadratic model?	Heights 5, 69, 101 and next 101 (t = 3): first differences 64, 32, 0; second differences −32, −32. Constant second differences mean quadratic.

Part 2: Claims, Mars & Models

#	Task	Answer
5	For what heights H does the equation $h(t) = H$ have two solutions, one solution, and no solutions? Use the discriminant.	$-16t^2 + 80t + (5 - H) = 0$: $D = 6400 + 64(5 - H) = 6720 - 64H$. Two solutions if $H < 105$, one if $H = 105$, none if $H > 105$.
6	On Mars, gravity is weaker: $h(t) = -6.1t^2 + 80t + 5$ (same launch). Find the maximum height and compare it to Earth's.	$t = 80 \div 12.2 \approx 6.56$ s; max $\approx 5 + 6400 \div 24.4 \approx 267.3$ ft, about 2.5 times Earth's 105 ft.
7	Classify each model as linear, exponential, or quadratic and write an equation for each.	A: $y = 4x + 3$. B: $y = 3(2)^x$. C: $y = x^2 + 2x + 3$.
8	For large x, which model in the table eventually has the greatest values? Explain.	Model B (exponential) eventually exceeds both.

Part 3: Design Within the Rules

#	Task	Answer
9	Show that the maximum height of $h(t) = -16t^2 + vt + 4$ is $4 + v^2 \div 64$.	Vertex at $t = v \div 32$: $h = -16(v \div 32)^2 + v(v \div 32) + 4 = -v^2 \div 64 + v^2 \div 32 + 4 = 4 + v^2 \div 64$.
10	Write and solve a compound inequality for the speeds v that follow the rule.	$80 \le 4 + v^2 \div 64 \le 100 \rightarrow 4864 \le v^2 \le 6144 \rightarrow$ about $69.7 \le v \le 78.4$ ft per s
11	Choose a speed, find its exact landing time, and present the launch with a graph and every key feature.	Example $v = 72$: max 85 ft; landing $t = (72 + \sqrt{5440}) \div 32 = (72 + 8\sqrt{85}) \div 32 = 2.25 + \frac{1}{4}\sqrt{85} \approx 4.55$ s

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Designed launch meets the height rule, with the vertex and zeros found algebraically. Goes beyond the minimum with a creative, realistic choice.	Designed launch meets the height rule, with the vertex and zeros found algebraically.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Launch report shows the function, a labeled graph, maximum height, landing time, and a check of the park rule. Polished, organized, and visually engaging.	Launch report shows the function, a labeled graph, maximum height, landing time, and a check of the park rule. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.